AI Agent SDKs & Frameworks: The Master Course (2026 Edition) 🎓🚀

Welcome to the definitive guide on **AI Agent SDKs and Frameworks**. This course is designed to transition you from "Chatting with AI" to **"Orchestrating Autonomous Business Systems."**

🏛️ Course Overview: The Age of Orchestration

The primary problem in modern software is no longer "Intelligence"—it is **Orchestration.**

If you ask a raw LLM to conduct a legal audit or a financial tax summary, it will likely hallucinate mathematical results or miss critical context. To solve this, we use **Agent Frameworks.**

In this course, we explore the major SDKs, solve **real-world industry problems** in Marketing, Finance, Legal, HR, and Real Estate, provide **interactive premium notebooks**, and evaluate the strengths of each framework.

🏗️ Chapter 1: The Microsoft Ecosystem

Microsoft caters to enterprise collaboration and deep architectural integration.

1. AutoGen: The Collaborative Swarm 🐝

- **The Business Problem**: A Q3 investment report for MSFT and TSLA requires data fetching, analysis, QA testing, and risk scoring — typically 4 different roles.
- **The SDK Solution**: AutoGen creates a 5-agent boardroom (CEO, Data Engineer, Analyst, QA Tester, Risk Manager) that autonomously debates and produces a verified report.
- **Key Features**: `GroupChat` orchestration, `%%capture` silent execution, live `yfinance` data, premium boardroom UI
- 📁 **Code**: `Chapter1_AutoGen_Company.ipynb`

2. Semantic Kernel: Enterprise AI Glue 🏗️

- **The Business Problem**: Raw LLM output must pass through brand rules (title case, 120-char limit, legal footer) before publication.
- **The SDK Solution**: Semantic Kernel wraps Python business logic as reusable **Native Plugins** via `@kernel_function`.
- **Key Features**: `BrandCompliancePlugin`, `ChatHistory` with strict prompts, green-on-black terminal UI
- 📁 **Code**: `Chapter1_SemanticKernel_Marketing.ipynb`

🕸️ Chapter 2: The LangChain Ecosystem

The industry standard for document intelligence and stateful logic.

1. LangChain: Retrieval-Augmented Generation (RAG) 📚

- **The Business Problem**: HR teams answer the same 50 questions daily. Employees wait hours for PTO, parental leave, and benefits answers.
- **The SDK Solution**: A full RAG pipeline: 7-section HR Handbook → RecursiveCharacterTextSplitter → OpenAI Embeddings → ChromaDB → Retrieval Chain.
- **Key Features**: 7 policy sections, 4 employee queries, Policy ID citations, dark-mode chat UI
- 📁 **Code**: `Chapter2_LangChain_HR_RAG.ipynb`

2. LangGraph: Deterministic State Machines 🤖

- **The Business Problem**: Support tickets misrouted 40% of the time. No sentiment awareness. VIPs treated same as standard users.
- **The SDK Solution**: A 7-node LangGraph state machine: Sentiment → Classifier → Priority → Handler → Escalation.
- **Key Features**: 4 intent types (refund/billing/technical/account), VIP detection, P0-P3 priority scoring, 3 test tickets
- 📁 **Code**: `Chapter2_LangGraph_Support.ipynb`

☁️ Chapter 3: Native Provider Tooling (Google & OpenAI)

Highly optimized agent SDKs built directly by the model creators.

1. Google Gen AI SDK (Agent Development Kit) 🔎

- **The Business Problem**: Real estate investors need factually grounded market intelligence across multiple markets — not AI hallucinations.
- **The SDK Solution**: Gemini 2.0 with **Search Grounding** — verifies every claim against live Google Search results.
- **Key Features**: 3-market comparison (Silver Lake / Austin / Miami), buy/hold/sell verdicts, luxury editorial UI
- 📁 **Code**: `Chapter3_GoogleADK_PropTech.ipynb`

2. OpenAI Assistants SDK 💪

- **The Business Problem**: Legal teams manually review vendor contracts. A missed liability clause can cost millions in litigation.
- **The SDK Solution**: Assistants API with **Stateful Threads** — audits 3 contract clauses with CRITICAL/HIGH/MEDIUM risk scoring.
- **Key Features**: 3-clause audit (Liability, Privacy, Termination), GDPR/CCPA citations, risk-colored legal report UI
- 📁 **Code**: `Chapter3_OpenAI_LegalAuditor.ipynb`

🧪 Chapter 4: Specialized Open-Source

Frameworks that solve role-playing and code-generation paradigms.

1. CrewAI: Role-Playing Collaboration 👥

- **The Business Problem**: 360° marketing campaigns require 4 different hires: researcher, strategist, copywriter, and brand reviewer.
- **The SDK Solution**: CrewAI orchestrates a 4-agent sequential pipeline where each agent builds on the previous output.
- **Key Features**: Rich agent backstories, FTC compliance review, 4-panel strategy deck UI
- 📁 **Code**: `Chapter4_CrewAI_Marketing.ipynb`

2. HuggingFace smolagents: Code-Native Execution 🧮

- **The Business Problem**: LLMs hallucinate numbers. Tax audits across UK, US, and EU require 100% mathematical precision.
- **The SDK Solution**: smolagents writes and executes Python code autonomously — 5 transactions, 2 discrepancies detected ($285 total).
- **Key Features**: Multi-region tax audit table, PASS/DISCREPANCY flags, deterministic Python execution
- 📁 **Code**: `Chapter4_SmolAgents_TaxAuditor.ipynb`

⚡ Chapter 5: Enterprise Low-Code Platforms

Scaling AI through orchestration platforms.

1. Dify: The Headless Agent Engine 🤖

- **The Business Case**: **Headless Sales Concierge.** Decouple agent logic from Python code so PMs can update the bot visually in Dify while engineers consume it as a REST API.
- **Key Features**: Stateful `conversation_id` tracking, blocking response mode, Enterprise tier pricing intelligence
- 📁 **Code**: `Chapter5_Dify_HeadlessAPI.ipynb`

2. n8n: The Autonomous Action Layer ⚙️

- **The Business Case**: **VIP Incident Escalation.** When a P0 VIP outage is detected, the agent fires a webhook to n8n triggering Slack alerts + CRM updates + priority support tickets.

- **Key Features**: JSON webhook payloads, P0_EMERGENCY priority detection, multi-app orchestration across 500+ integrations
- 📁 **Code**: `Chapter5_n8n_ActionOrchestrator.ipynb`

🔬 Chapter 6: Observability (Production Guardrails)

Building an agent is easy; knowing *why* it failed is hard.

The "Black Box" Problem

In production, you need **Tracing**. Tools like **LangSmith** and **Arize Phoenix** act as the "Datadog" of AI, giving you a waterfall view of every tool call, token count, and latency spike.

⚡ Interactive Learning Environment

Every chapter includes a **Premium Interactive Notebook** that you can run locally: * **Visual UI Contexts**: Every agent demo features a custom-designed, high-fidelity UI rendered directly in your notebook (dark terminals, editorial layouts, strategy decks). * **Cost & Speed Optimized**: All notebooks use `gpt-4o-mini` by default (0.15¢ per 1M tokens) to ensure your learning is virtually free. * **Hybrid Simulation Mode**: No API Keys? No problem. Every notebook features a high-fidelity simulation mode so you can see the agentic behavior regardless of your environment. * **Silent Execution**: `%%capture` magic hides all framework logs — output is rendered via premium HTML/CSS UIs only.

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